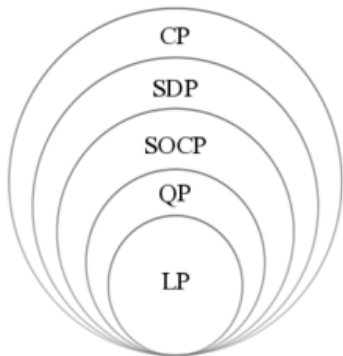
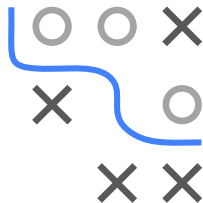


Optimization in Machine Learning

Constrained Optimization Introduction



Learning goals

- Examples of constrained optimization in statistics and ML
- General definition
- Hierarchy of convex constrained problems

CONSTRAINED OPTIMIZATION IN STATISTICS

Example: Maximum Likelihood Estimation

- For data $(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)})$, we want to find the MLE

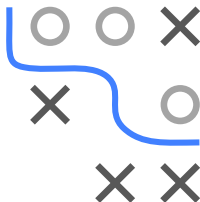
$$\max_{\theta} L(\theta) = \prod_{i=1}^n f(\mathbf{x}^{(i)}, \theta)$$

- In some cases, θ can only take **certain values**
- If f is a Poisson distribution, we require $\lambda \geq 0$
- If f is a multinomial distribution

$$f(x_1, \dots, x_p; n; \theta_1, \dots, \theta_p) = \begin{cases} \frac{n!}{x_1! \dots x_p!} \theta_1^{x_1} \dots \theta_p^{x_p} & \text{if } x_1 + \dots + x_p = n \\ 0 & \text{else} \end{cases}$$

- The probabilities θ_i must lie between 0 and 1 and sum to 1:

$$0 \leq \theta_i \leq 1 \text{ for all } i \quad \text{and} \quad \theta_1 + \dots + \theta_p = 1$$



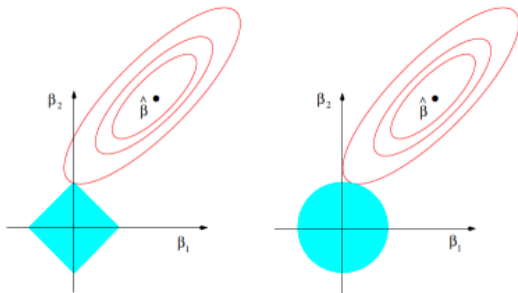
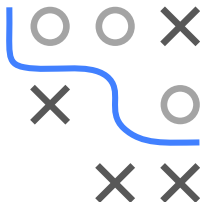
CONSTRAINED OPTIMIZATION IN ML

- **Lasso regression:**

$$\min_{\beta \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^n \left(y^{(i)} - \beta^T \mathbf{x}^{(i)} \right)^2 \quad \text{s.t. } \|\beta\|_1 \leq t$$

- **Ridge regression:**

$$\min_{\beta \in \mathbb{R}^p} \frac{1}{n} \sum_{i=1}^n \left(y^{(i)} - \beta^T \mathbf{x}^{(i)} \right)^2 \quad \text{s.t. } \|\beta\|_2 \leq t$$

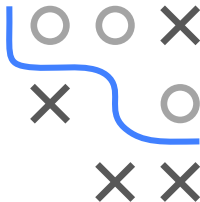


CONSTRAINED OPTIMIZATION IN ML

- Dual formulation of the SVM: convex QP with box constraints plus one linear constraint

$$\max_{\alpha \in \mathbb{R}^n} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y^{(i)} y^{(j)} \langle \mathbf{x}^{(i)}, \mathbf{x}^{(j)} \rangle$$

$$\text{s.t. } 0 \leq \alpha_i \leq C, \quad \sum_{i=1}^n \alpha_i y^{(i)} = 0$$



CONSTRAINED OPTIMIZATION

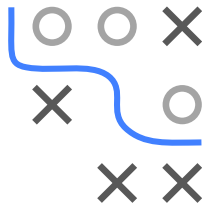
- **Constrained Optimization problem:**

$$\min f(\mathbf{x}) \quad \text{s.t. } g_i(\mathbf{x}) \leq 0 \ (i = 1, \dots, k), \quad h_j(\mathbf{x}) = 0 \ (j = 1, \dots, l)$$

- $g_i : \mathbb{R}^d \rightarrow \mathbb{R}$ are **inequality constraints**
- $h_j : \mathbb{R}^d \rightarrow \mathbb{R}$ are **equality constraints**

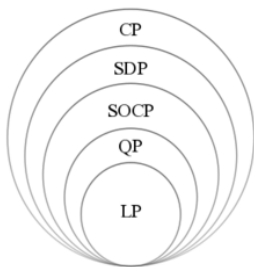
- The **feasible set** is the set of inputs $\mathbf{x}$ that fulfill the constraints:

$$\mathcal{S} := \{\mathbf{x} \in \mathbb{R}^d \mid g_i(\mathbf{x}) \leq 0, h_j(\mathbf{x}) = 0 \ \forall \ i, j\}$$



CONSTRAINED CONVEX OPTIMIZATION

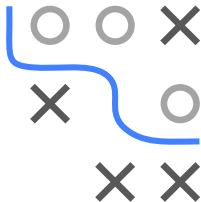
- **Convex programs:** convex objective f , convex inequality constraints g_i , and affine equality constraints h_j (i.e., $h_j(\mathbf{x}) = \mathbf{A}_j^T \mathbf{x} - \mathbf{b}_j$)



- **Linear program (LP):** f and all constraints g_i, h_j are linear
- **Quadratic program (QP):** f is a quadratic form, constraints are linear

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{c}^T \mathbf{x} + d \quad \text{for } \mathbf{Q} \in \mathbb{R}^{d \times d}, \mathbf{c} \in \mathbb{R}^d, d \in \mathbb{R}$$

- Also: second-order cone programs (SOCP), semidefinite programs (SDP), cone programs (CP)



CONSTRAINED CONVEX OPTIMIZATION

- SOCPs play a pivotal role in statistics and engineering
 - Lobo et al. 1998
- In ML, SDPs are at the heart of, e.g., learning kernels from data
 - Lanckriet et al. 2004
- This categorization of convex optimization problem classes helps design specialized *optimization methods* tailored to specific problem types
- Keyword: disciplined convex programming
 - Grant, Boyd, and Ye 2006

